IMO 2026 — Problem 6

Solution by GPT-5.6 Sol (max reasoning)

2026-07-22

Source: `results/gpt-5.6-sol-max/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $a_1, a_2, a_3, \ldots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n, the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \ldots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n. (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y.)

Solution document

Status

partial

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Approaches tried

- Recast the condition using prime-divisor sets. Define $S_n = \{x > 1 : \gcd(x, a_i) > 1 \text{ for all } i \leq n\}$. Then $S_{n+1} \subseteq S_n$, and a_{n+1} is the least member of S_n above a_n.
- Identified a sufficient stabilization criterion: if S_N is pairwise non-coprime, then every later term lies in S_N and imposes no new restriction on it, so $S_n = S_N$ thereafter. The sequence is then the increasing enumeration, starting with a_1, of a periodic set of integers.
- Established the bounded-gap observation: every multiple of a_1 belongs to every S_n, hence $1 \leq a_{n+1} - a_n \leq a_1$.

- Explored finite prime supports and computed examples. For $a_1 = 15$, the sequence starts $15, 18, 20, 24, 30, 36, 40, 42, 45, 48, 50, 54, 60, \ldots$ and has $(T, L) = (8, 30)$. After the first three supports $\{3, 5\}, \{2, 3\}, \{2, 5\}$, the stable admissible set is the set of numbers divisible by at least two of $2, 3, 5$.
- The simple hope that a stable admissible set must be a single divisibility class is false; the $a_1 = 15$ example shows that a finite union of classes is needed.

Current best

It remains to prove finite stabilization. The problem has been reduced to a combinatorial/arithmetic assertion about the descending admissible sets S_n. Each S_n is periodic (its membership depends only on divisibility by the finitely many primes occurring in $a_1, \ldots, a_n$), but new primes may enter later terms. A likely route is to use the bounded gaps together with the fact that every term has a prime factor from the fixed finite set $P(a_1)$, or to formulate the process as successively adjoining hyperedges and prove that it reaches a pairwise-intersecting blocker after finitely many steps. Once stabilization is proved, periodicity of the sequence, including the identity from $n = 1$, can be derived by choosing a common period divisible by all relevant primes and checking the counting shift.

Full proof

Not yet complete. Established pieces are as follows.

Let

$$S_n = \{x > 1 : \gcd(x, a_i) > 1 \quad (1 \le i \le n)\}.$$

Every multiple of a_1 lies in S_n, since each a_i has gcd greater than 1 with a_1. Thus the recurrence is always defined and

$$a_n < a_{n+1} \le a_n + a_1.$$

If for some N every two members of S_N have gcd greater than 1, then all future terms belong to S_N. Each such term is non-coprime to every member of S_N, so adding it as a constraint does not shrink S_N. Hence $S_n = S_N$ for all $n \ge N$, and the recurrence successively lists every member of S_N above the current term. Since S_N is defined by finitely many gcd conditions, it is periodic modulo the product of the finitely many primes dividing $a_1 \cdots a_N$. This gives a constant index shift under translation by that modulus. The remaining issue is to prove that such an N exists and then make the all-n indexing argument fully explicit.